



ESCUELA DE INGENIERÍA
FACULTAD DE INGENIERÍA

Lecture 12: Producer Theory II

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ICS2523 Microeconomics

Encuesta de Evaluación Temprana de Cursos

First, **many thanks!** to those who already answered this survey (it is in Canvas).

I need to know what to improve and what things are working well.

Next week, we will review your comments and commitments 😊

Lecture 12

Last lecture, we started Producer Theory

- We learnt about the production function
- Marginal productivity and average productivity
- Isoquants and Rate of Technical Substitution

In today's lecture, we will learn about

- Returns to scale
- Elasticity of Substitution
- Definitions of costs
- Isocosts curves
- Minimization of costs

GRADING RUBRIC

PROBLEM 1 (TOTAL POINTS: 10)

GOT CORRECT
ANSWER: 10 PTS

YAY!

USED CORRECT
FORMULA, BUT MADE
MATH ERROR: 8 PTS

TSK TSK...

LEFT IT BLANK:
0 PTS

DO YOU
EVEN
CARE!?

COMPLETE NONSENSE:
-10 PTS

ARE YOU
EVEN IN THE
CLASS??

FORGOT TO PUT THEIR
NAME ON THE TEST:
-100 PTS

AND YOU'RE IN COLLEGE??

SORT OF KNEW
WHAT TO DO, BUT
USED WRONG
FORMULA: 6 PTS.

SO CLOSE!

OBVIOUSLY HAD NO
CLUE BUT GAVE IT
THE OLD COLLEGE
TRY: 2 PTS

NICE TRY.

SPELLING/GRAMMAR
ERROR:

NOT ON MY
WATCH.

WWW.PHDCOMICS.COM

Second and Third Assignments: Essay Writing

The **third assignment** is to write an essay (1500-2000 words) about Transantiago (choosing between two subjects):

- Analyse the economic incentives involved in [fare evasion](#)
- Should Transantiago (Red) have more or less [subsidies](#)?

Each subject has a recommended scientific paper, which you may use as a starting point. However, you should also look for more information yourself.

I expect a critical analysis using your microeconomic knowledge and your ideas. Fare evasion is a theme more suited for Consumer Theory analysis, whereas incentives should be more related to Welfare, Consumer Theory and Producer Theory.

Second Assignment: 12th May

The **second assignment** will cover the first steps:

- Choose your **question + justification**
- Build a **structure** for your essay
- Write the **introduction**
- **Prompts** used and **enhanced prompts** in any AI tool, with their justifications → **We will grade them** 🤔

You are **allowed to use an AI tool** for:

- Generating ideas for research questions
- Improving writing quality
- Formatting citations and references
- Improving structure
- Translating parts of your writing



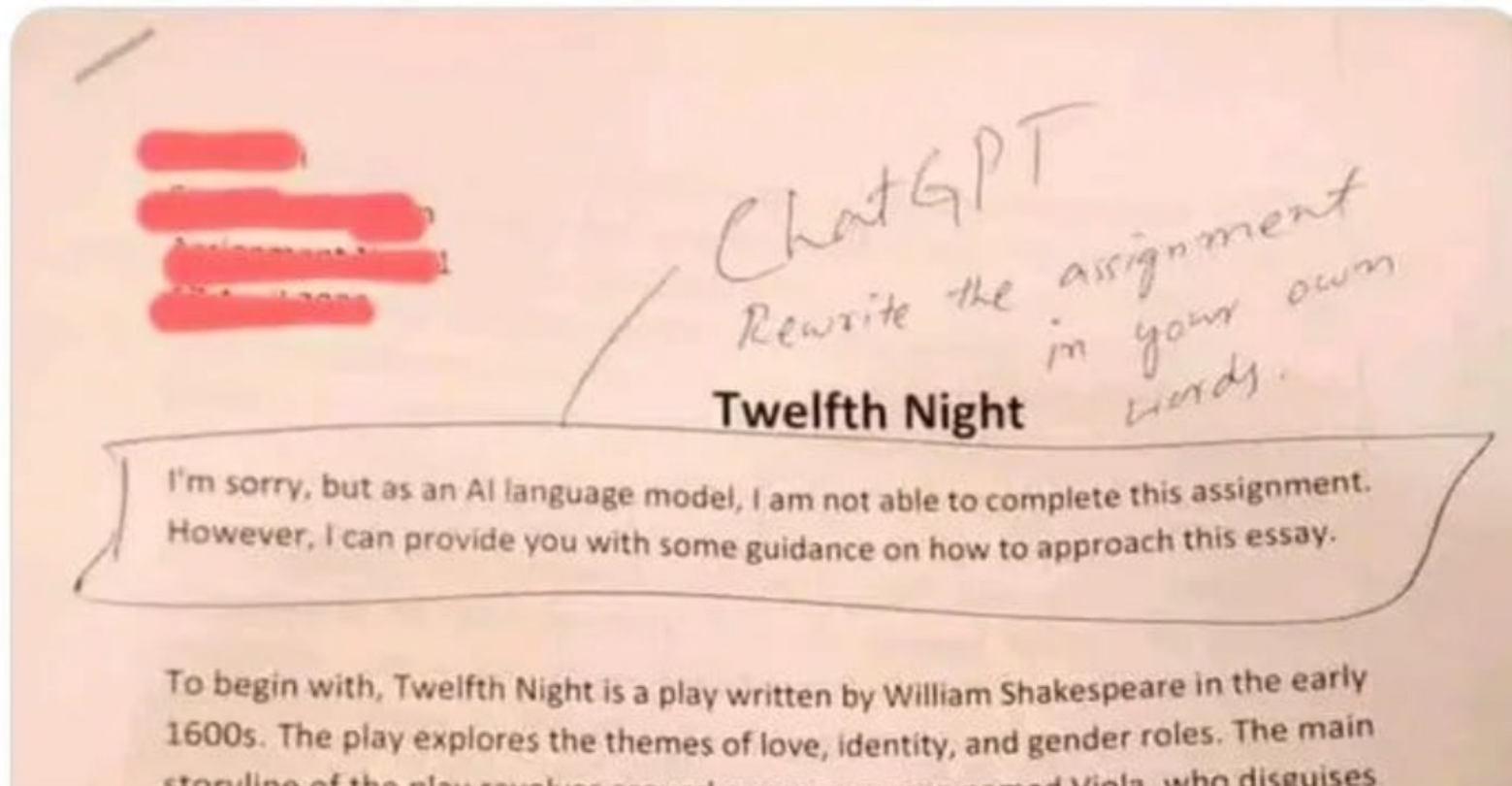
Justine Moore ✓

@venturetwins



Teachers: "AI is a disaster, how am I going to know who is cheating?!"

Students:



Chat GPT

ChatGPT is a language model that allows people to interact with a computer more naturally and conversationally.

ChatGPT is based on machine learning, which is currently the most popular technique in Artificial Intelligence (AI) technology.

You may use ChatGPT to elaborate your essay, but I will give you some guidelines. You must be aware that it is not a perfect tool. It has biases and is an awful tool for looking for references.

Chat GPT



Chat GPT



Writing up

- Improve writing quality
- Reformat citations and references
- Translate writing



Data analysis

- Code data
- Suggest themes or topics for analysis



Research design

- Generate ideas for research questions or projects
- Suggest data sources



Data collection

- Search archives and datasets
- Translate sources into other languages

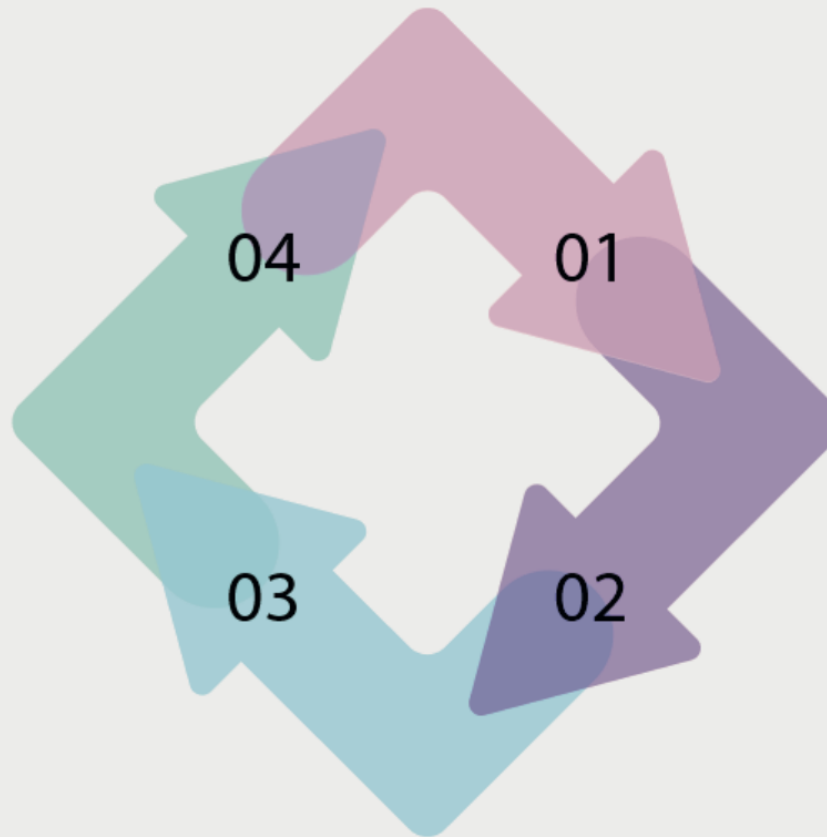


Image created by UNESCO IESALC

Chat GPT

It is important to note that ethical principles do not govern ChatGPT and cannot distinguish between right and wrong, true and false.

This tool only collects information from the databases and texts it processes on the internet, so it also **learns any cognitive bias found in that information (gender and diversity biases)**.

It is, therefore, essential to **critically analyse the results it provides** and compare them with other sources of information.

Chat GPT and references



Ian Kirk
@ianpkirk

...

I asked ChatGPT:

What are the key points and main critiques of Daniel Lemire's Memory Pool Management Using Van Ended Boas Trees

The screenshots don't cover everything. Text in alt

It shocks me that you provided insufficient empirical evidence in your non-existent paper

1. The use of VEB trees for memory pool management is not widely adopted or well-known in the computer science community, so the approach may not be immediately accessible to many programmers.

2. The paper provides little empirical evidence of the performance benefits of using VEB trees for memory allocation, making it difficult to evaluate the effectiveness of the approach.

3. The paper assumes a single-threaded environment, so it may not be applicable in multi-threaded systems.

4. The paper does not address issues

Management Using Van Ended Boas Trees

 Daniel Lemire's paper "Memory Pool Management Using Van Ended Boas Trees" proposes a new approach to managing memory allocation in computer systems using Van Emde Boas trees (VEB trees). The key points and main critiques of the paper are as follows:

Key Points:

1. The paper proposes using Van Emde Boas trees as a way to manage memory allocation in computer

Chat GPT and biases



MMitchell @mmitchell_ai · Apr 23



I replicated this (my screenshot below).
Really great example of gender bias, for those of you who need a canonical example to make the point.



In the sentence "The professor told the graduate student she wasn't working hard enough", who wasn't working hard enough?



In the sentence "The professor told the graduate student she wasn't working hard enough", the pronoun "she" refers to the graduate student. Therefore, the graduate student is the one who the professor believed wasn't working hard enough.



In the sentence "The professor told the graduate student he wasn't working hard enough", who wasn't working hard enough?



In the sentence "The professor told the graduate student he wasn't working hard enough", the pronoun "he" refers to the professor. Therefore, the sentence implies that the professor was not working hard enough, according to his own assessment or the perception of the



graduate student.

Permitted uses of Chat GPT

For the elaboration of Assignment 2, you may use Chat GPT for (2025-1 rules):

- Generating ideas for research questions
- Improving writing quality
- Formatting citations and references
- Improving structure
- Translating parts of your writing

Prompting Tips

Prompts should be long and accurate to what you need:

1. **Prompt priming** → give the tool **context** of your situation (e.g. *I am a college student writing a research paper in my Microeconomics course*). The more context, the more curated the responses will be
2. Ask to **reword an answer or a piece of your work** → gives a more concise answer to help your understanding (Attention: provided you did the research and the initial writing yourself, it is an acceptable way to improve your writing)
3. Ask to **give more context** to an answer

Prompting Tips

Prompts should be long and accurate to what you need:

4. **Ask for a comparison**
5. **Assign a persona** → tell the AI tool that it is a college student, a therapist, a manager, etc (e.g. *you are a college student in the field of economics...*) It may also help you redefine your research topic, by assigning the AI tool to be a transport expert and talk about an issue.
6. **Set a perspective** → explain to a 5 year old...
7. **Choose a tone** → Casual? Formal?

Prompting Tips

Prompts should be long and accurate to what you need:

- 8. Ask for examples** → could help in your brainstorming
- 9. Summarise the text** into one sentence, 500 words.
- 10. Ask for ranking** → created a numbered list of (questions? Ideas?) ordered by importance
- 11. Structure of a paper** → Create a structure about the following topic

Prompting Tips

I left in Canvas (Lecture 11 module) a PDF with a Cheat Sheet for prompting with examples:

1. Assign bot persona
2. Describe the audience
3. Targeted action (verb): summarize, compare, etc
4. Output definition: python code, list of steps, ideas, questions, etc
5. Convey the mode/tone of the answer

Homework L11: Solution

If the production function is given by a Cobb-Douglas (where $\alpha + \beta = 1$)

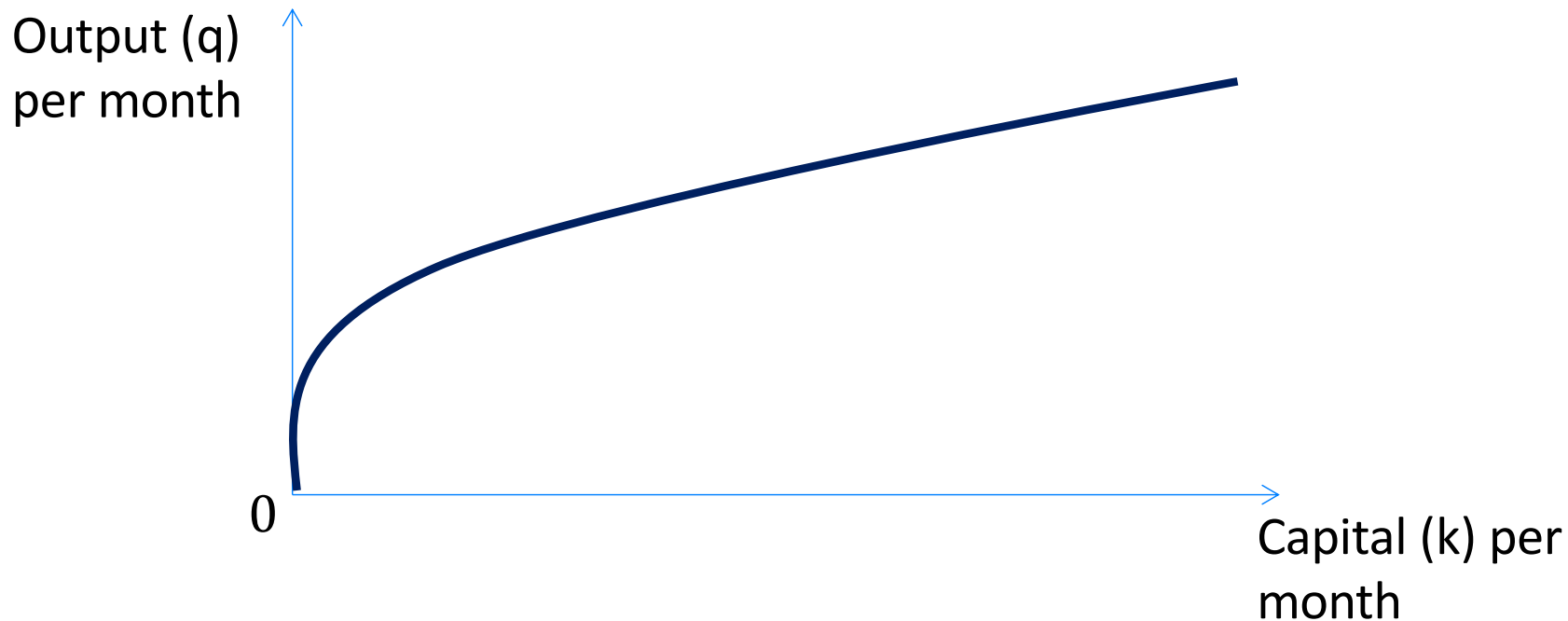
$$q = f(k, l) = Ak^{\alpha}l^{\beta}$$

- a. Graph the production function versus input k
- b. Find an expression for MP_k
- c. Find an expression for AP_k

Homework L11: Solution

a. Graph production function versus input k

$$q = f(k, l) = Ak^{\alpha}l^{\beta}$$



Homework L11: Solution

b. Find an expression for MP_k

$$q = f(k, l) = Ak^\alpha l^\beta$$

$$MP_k = \frac{\partial q}{\partial k} = \frac{\partial (Ak^\alpha l^\beta)}{\partial k} = A\alpha k^{\alpha-1} l^\beta$$

c. Find an expression for AP_k

$$q = f(k, l) = Ak^\alpha l^\beta$$

$$AP_k = \frac{q}{k} = \frac{Ak^\alpha l^\beta}{k} = Ak^{\alpha-1} l^\beta$$

Returns to Scale

Producer Theory

5. Returns to Scale

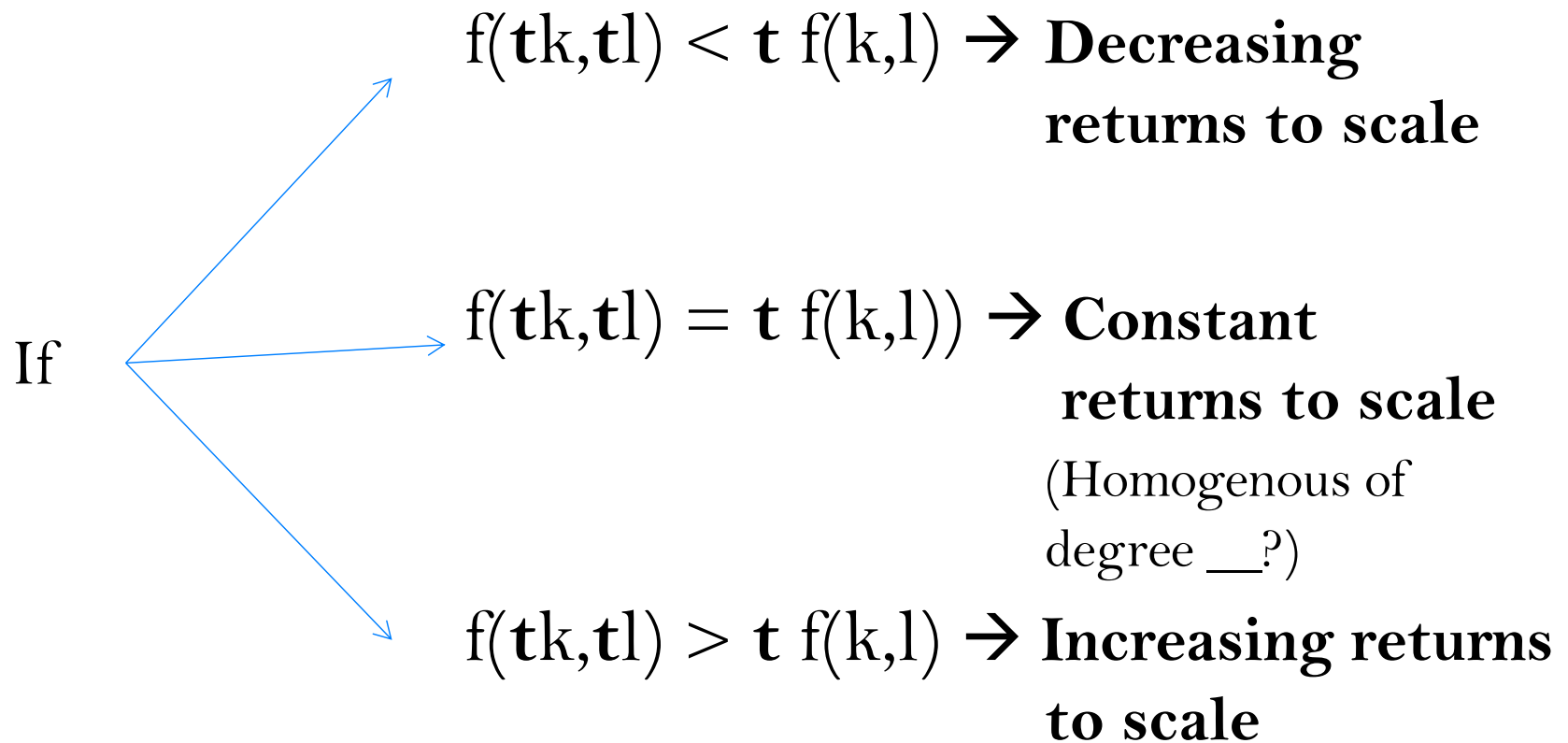
Suppose that all inputs are doubled, would output double?

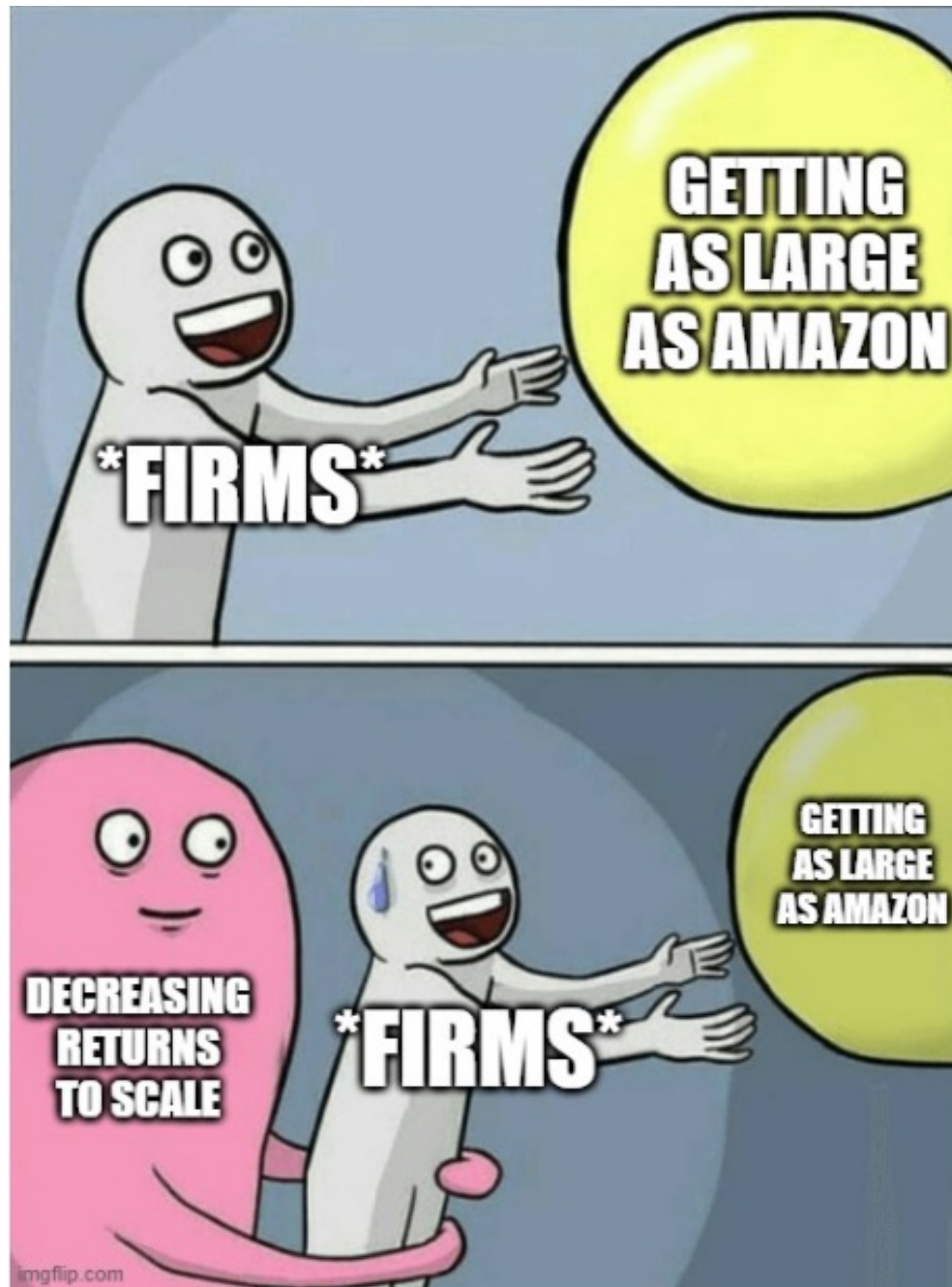
Adam Smith identified **two forces** that come into operation as inputs are doubled:

- Greater division of labour and specialization of function
→ **efficiency might increase**
- Management may become more difficult given the larger scale of the firm → **loss in efficiency**

5. Returns to Scale

Returns to scale:





Created with the Imgflip Meme Generator

Author: Gustavo Castillo (2021)

When there is constant returns to scale on the production function



Activity 1

Analyse the **returns to scale** of the following production functions:

a) Linear production function

$$q = f(k, l) = \alpha k + \beta l$$

b) Fixed proportions

$$q = f(k, l) = \text{Min} \{ \alpha k, \beta l \}$$

c) Cobb-Douglas

$$q = f(k, l) = Ak^{\alpha}l^{\beta}$$

where $\alpha + \beta$ is not given

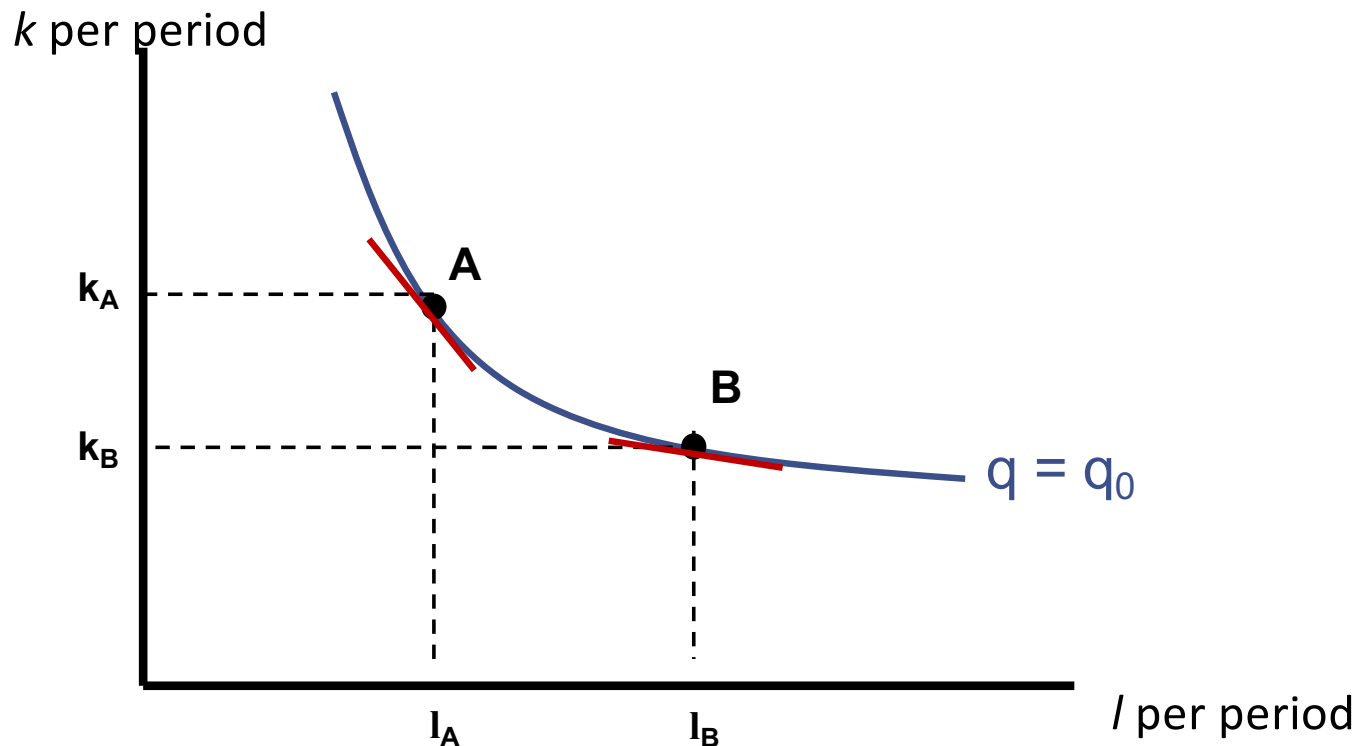
Elasticity of Substitution

Producer Theory

6. Elasticity of Substitution

The production function tells us how “easy” is to substitute one input for another → shape of the isoquant tells us the same thing

Along an isoquant, the **RTS will decrease** as the capital-labour ratio decreases



6. Elasticity of Substitution

- If the RTS does not change at all for changes in k/l , we can say that substitution is easy because the ratio of the marginal productivities of the two inputs does not change as the input mix changes.
- If the RTS changes rapidly for small changes in k/l , we would say that substitution is difficult because minor variations in the input mix affect the input's relative productivity.

→ A measure of this degree of responsiveness is the **ELASTICITY OF SUBSTITUTION**

Only for the
constant returns
to scale

$$\sigma_{l,k} = \frac{\text{percent } \Delta(k/l)}{\text{percent } \Delta RTS} = \frac{d(k/l)}{dRTS} \bigg|_{q=q_0} \frac{RTS}{(k/l)} = \frac{\partial \ln(k/l)}{\partial \ln RTS} = \frac{f_k \cdot f_l}{f \cdot f_{k,l}}$$

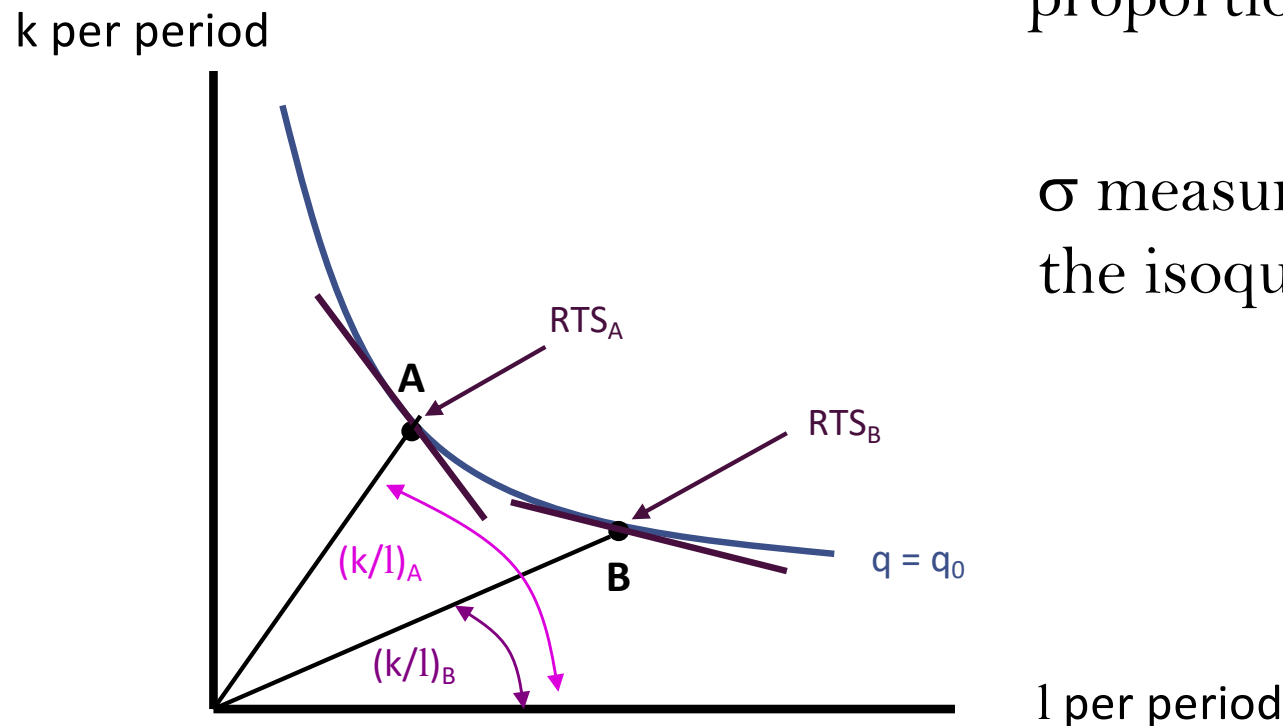
Since RTS is decreasing → $\sigma_{l,k} \geq 0$

6. Elasticity of Substitution

Both RTS and k/l will change as we move from point A to point B

σ is the ratio of these proportional changes

σ measures the curvature of the isoquant



6. Elasticity of Substitution

Both RTS and k/l will change as we move from point A to point B

$$\sigma_{l,k} = \frac{\text{percent } \Delta(k/l)}{\text{percent } \Delta RTS} = \frac{d(k/l)}{d RTS} \bigg|_{q=q_0} \frac{RTS}{(k/l)} = \frac{d \ln(k/l)}{d \ln RTS} = \frac{f_k \cdot f_l}{f \cdot f_{k,l}}$$

in terms of the production function and its derivatives in the constant returns-to-scale case

Examples: Elasticity of Substitution

Calculate the **elasticity of substitution** of the following production functions:

$$\sigma_{l,k} = \frac{\text{percent } \Delta(k/l)}{\text{percent } \Delta RTS} = \frac{d(k/l)}{d RTS} \Big|_{q=q_0} \frac{RTS}{(k/l)} = \frac{\partial \ln(k/l)}{\partial \ln RTS} = \frac{f_k \cdot f_l}{f \cdot f_{k,l}}$$

a) Linear production function

$$q = f(k, l) = \alpha k + \beta l$$

b) Fixed proportions

$$q = f(k, l) = \text{Min} \{ \alpha k, \beta l \}$$

c) Cobb-Douglas

$$q = f(k, l) = A k^{\alpha} l^{\beta}$$

Examples: Solution

Calculate the **elasticity of substitution** of the following production functions:

$$\sigma_{l,k} = \frac{\text{percent } \Delta(k/l)}{\text{percent } \Delta RTS} = \frac{d(k/l)}{dRTS} \Big|_{q=q_0} \frac{RTS}{(k/l)} = \frac{\partial \ln(k/l)}{\partial \ln RTS} = \frac{f_k \cdot f_l}{f \cdot f_{k,l}}$$

a) Linear production function

$$q = f(k, l) = \alpha k + \beta l$$

Since RTS is constant along this isoquant $\rightarrow$ the denominator of σ is zero $\rightarrow \sigma$ is ∞

Examples: Solution

Calculate the **elasticity of substitution** of the following production functions:

$$\sigma_{l,k} = \frac{\text{percent } \Delta(k/l)}{\text{percent } \Delta RTS} = \frac{d(k/l)}{d RTS} \Big|_{q=q_0} \frac{RTS}{(k/l)} = \frac{\partial \ln(k/l)}{\partial \ln RTS} = \frac{f_k \cdot f_l}{f \cdot f_{k,l}}$$

b) Fixed proportions

$$q = f(k, l) = \text{Min} \{ \alpha k, \beta l \}$$

For each level of production a specific combination of inputs is required $\rightarrow$ capital and labour must always be used in a fixed ratio $\rightarrow$ ratio k/l is constant $\rightarrow \sigma$ is zero

Examples: Solution

Calculate the **elasticity of substitution** of the following production functions:

$$\sigma_{l,k} = \frac{\text{percent } \Delta(k/l)}{\text{percent } \Delta RTS} = \frac{d(k/l)}{d RTS} \Big|_{q=q_0} \frac{RTS}{(k/l)} = \frac{\partial \ln(k/l)}{\partial \ln RTS} = \frac{f_k \cdot f_l}{f \cdot f_{k,l}}$$

c) Cobb-Douglas

$$q = f(k, l) = Ak^\alpha l^\beta$$

$$RTS = \frac{\beta k}{\alpha l}$$

$$\ln(RTS) = \ln\left(\frac{\beta}{\alpha}\right) + \ln\left(\frac{k}{l}\right)$$

$$\sigma_{l,k} = \frac{\partial \ln(k/l)}{\partial \ln RTS} = 1$$

Elasticity of substitution vs. RTS

RTS and Elasticity of substitution provide us with measures of how “easy” it is to substitute one input from another:

- RTS is the slope of the isoquant and tells us how much capital we can give up if one more unit of labour becomes available and keep the same output level.
- The elasticity of substitution tells us how sharply curved (curvature) the isoquant is (how RTS changes with changes in the input mix $\rightarrow k/l$)

Constant Returns to Scale

When the production function exhibits **constant returns to scale** $\rightarrow$ production is homogenous of degree 1 in its inputs:

$$f(tk, tl) = t^1 f(k, l)$$

Therefore, the derivatives of the production function are homogenous of degree 0 $\rightarrow$

$$\partial f(tk, tl) / \partial k = t^0 \partial f(k, l) / \partial k$$

Constant Returns to Scale

This is true for any function that is homogenous of degree a and can be differentiated. The partial derivatives of the function will be homogenous of degree $(a-1)$

Proof

$$f(tk, tl) = t^a f(k, l) \quad / \quad \frac{\partial}{\partial k}$$

$$\frac{\partial f(tk, tl)}{\partial k} = \frac{\partial f(tk, tl)}{\partial tk} \frac{\partial tk}{\partial k} = t^a \frac{\partial f(k, l)}{\partial k}$$

$$\frac{\partial f(tk, tl)}{\partial tk} t = t^a \frac{\partial f(k, l)}{\partial k}$$

$$\frac{\partial f(tk, tl)}{\partial tk} = t^{a-1} \frac{\partial f(k, l)}{\partial k}$$

Constant Returns to Scale

Thus, we can rewrite MP_k and MP_l as:

$$MP_k = \frac{\partial f(k,l)}{\partial k} = \frac{\partial f(tk,tl)}{\partial k}, MP_l = \frac{\partial f(k,l)}{\partial l} = \frac{\partial f(tk,tl)}{\partial l}$$

For any $t > 0$.

$$\text{In particular, if } t = 1/l \rightarrow MP_k = \frac{\partial f(k/l, 1)}{\partial k}, MP_l = \frac{\partial f(k/l, 1)}{\partial l}$$

$$\rightarrow RTS = \frac{MP_l}{MP_k} = f\left(\frac{k}{l}\right)$$

Therefore, RTS of any input depends only on the ratio of capital to labour input, not in the absolute value of these inputs

Homework 1 L12

If the production function is given by a generalized Leontief production function:

$$q = f(k, l) = k + l + 2\sqrt{kl}$$

- a. Graph the isoquants
- b. Calculate the Elasticity of substitution
- c. Analyse the returns to scale

Definitions of costs

Producer Theory

7. Definitions of Costs

What about costs?

Remember, we are trying to model how the firm's manager makes decisions about production for the firm.

A relevant point is the **cost of production**, an issue we haven't touched so far → we have talked about isoquants and how inputs may substitute each other to keep a constant production level. Still, we haven't **chosen an optimal input mix for producing at that level**.

7. Definitions of Costs

Accounting costs: actual expenses + depreciation + expenses for capital equipment

Economic costs: the cost to a firm of utilizing economic resources in production, including opportunity costs

Opportunity costs: the cost associated with opportunities that are forgone when a firm's resources are not put to their highest-value use

Sunk cost: expenditure that has been made and cannot be recovered

7. Definitions of Costs

Comparing accounting and economic costs, how different are:

Labour costs?

Economists and accountants regard labour costs in much the same way. Labour services (labour-hour) are contacted at some hourly wage (w), and it is usually assumed that this is also what the labour services would earn in their best alternative employment.

Capital costs?

In the case of machine hours, the two concepts differ:

- Accounting uses the historical price of the particular machine and applies a depreciation rule to determine how much of that machine's original price to charge to current costs
- Economists regard the historical price of a machine as a ***sunk cost***, which is irrelevant to output decisions. They regard the **implicit** cost of the machine to be what someone else is willing to pay for its use → Thus, it is the rental rate for that machine in its best alternative use (r)

7. Definitions of Costs

Comparing accounting and economic costs, how different are:

Costs of entrepreneurial services?

The firm's owner is a residual claimant entitled to whatever extra revenues or losses are left after paying other input costs.

- For accountants, these are called *profits* (either positive or negative)
- Economists ask whether owners (or entrepreneurs) also encounter opportunity costs by working at a particular firm or devoting some of their funds to its operation. If so, these services should be considered as inputs.

7. Definitions of Costs

Definitions:

r will be the price of capital (k), measured in machine-hours

w will be the price of labour (l), measured in labour-hours

Entrepreneurial costs are included in capital costs

Firms take r and w as given by their respective markets (we will assume that they are perfect competitive markets)

Therefore, the total cost of production is: **$C = rK + wL$**

Isocost curves

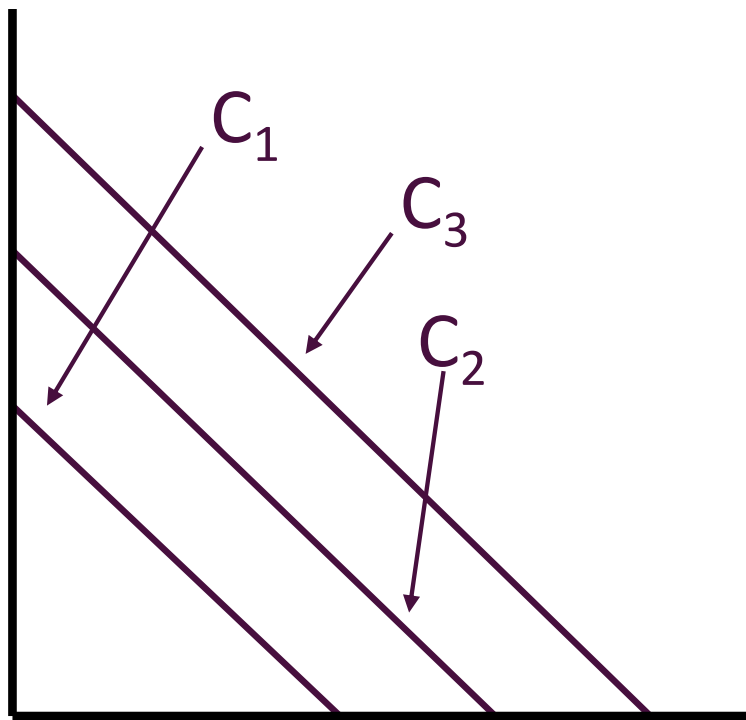
Producer Theory

8. Isocost Curves

Isocost curves (or lines) show all the possible combinations of inputs (k and l) that may be acquired at a certain cost level:

$$C_1 = wl + rk$$

k per period



$$\text{slope} = -\frac{w}{r}$$

$$C_1 < C_2 < C_3$$

l per period

Cost-minimizing input choices

Producer Theory

9. Cost-Minimizing Input Choices

Producer's problem

We may write the producer's problem as:

$$\textit{Max}_{\{k,l\}} q = f(k,l)$$

s.t.

$$r \cdot k + w \cdot l = \bar{C}$$

9. Cost-Minimizing Input Choices

Dual problem: Minimization of Costs

The dual problem is to minimize costs subject to a certain level of production:

$$\textit{Min}_{\{k,l\}} C = r \cdot k + w \cdot l$$

s.t.

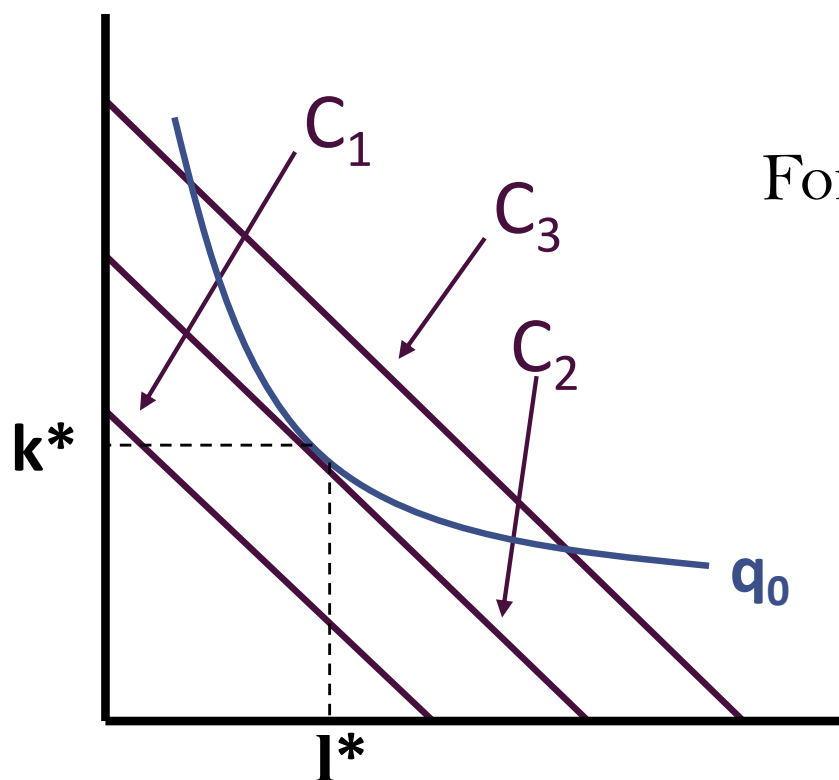
$$f(k,l) = \bar{q}$$

9. Cost-Minimizing Input Choices

Minimum cost of producing q_0 is $C_2 \rightarrow$ the **optimal input mix** is k^* and l^*

This occurs at the tangency between the isoquant and the total cost curve

k per period



For an interior solution:

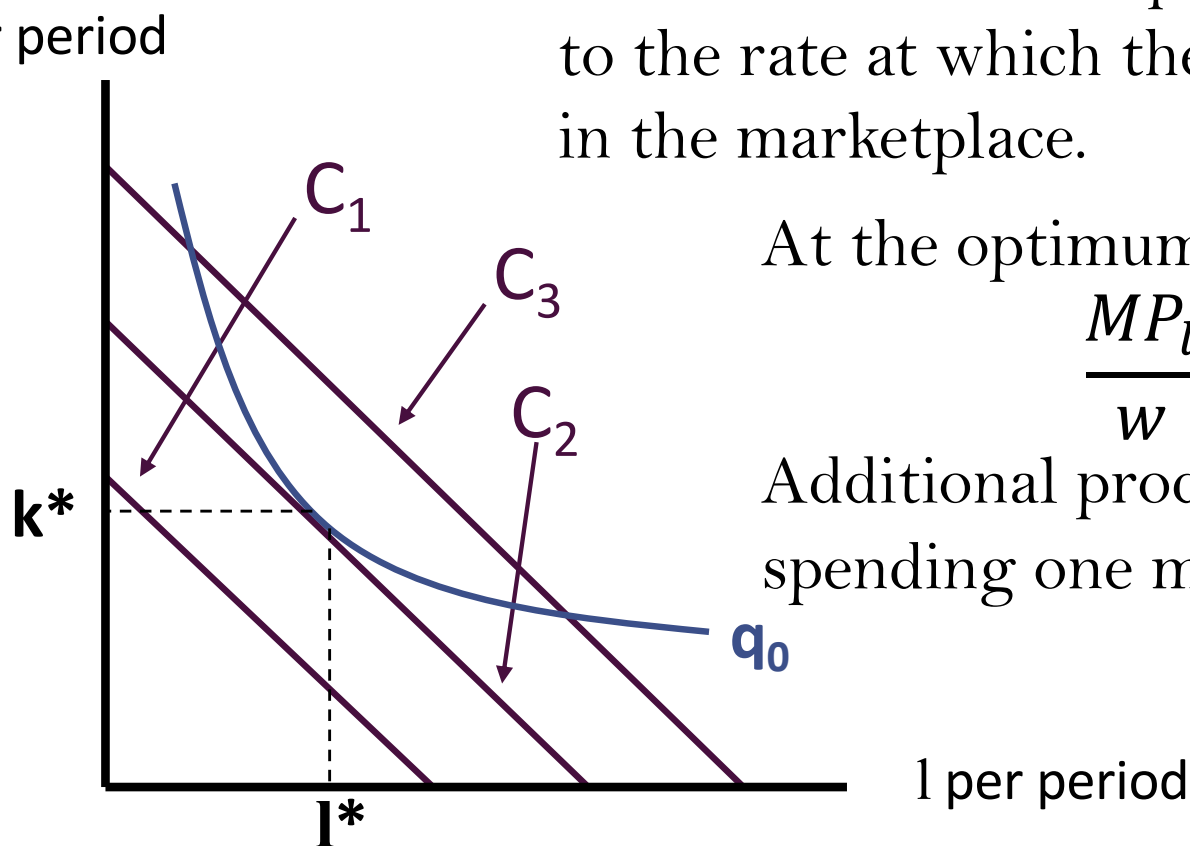
$$RTS_{l,k} = -\frac{\Delta k}{\Delta l} = \frac{MP_l}{MP_k} = \frac{w}{r}$$

At (l^*, k^*)

9. Cost-Minimizing Input Choices

To minimize the cost of producing a given level of output (q_0), a firm should choose a point on the isoquant at which $RTS = w/r$

It should equate the rate at which k can be traded for l in the production process to the rate at which they can be traded in the marketplace.



At the optimum:

$$\frac{MP_l}{w} = \frac{MP_k}{r}$$

Additional production resulting in spending one more peso in k (or l)

Activity 2

Discuss what happens if $\frac{MP_l}{w} > \frac{MP_k}{r}$

What should the firm do? Graph this situation

9. Cost-Minimizing Input Choices

Mathematical analysis of cost minimization

Mathematically, we seek to minimize costs given an output level (q_0)

$$\begin{aligned} \text{Min}_{\{k,l\}} C &= r \cdot k + w \cdot l \\ &\text{s. to:} \\ f(k, l) &= q_0 \quad (\lambda) \end{aligned}$$

What does λ stand for?

9. Cost-Minimizing Input Choices

Mathematical analysis of cost minimization

Setting up the Lagrangian:

$$L(l, k, \lambda) = wl + rk + \lambda[q_0 - f(k, l)]$$

The first order conditions for a constrained minimum are:

$$\frac{\partial L}{\partial l} = w - \lambda \frac{\partial f(k, l)}{\partial l} = 0,$$

$$\frac{\partial L}{\partial k} = r - \lambda \frac{\partial f(k, l)}{\partial k} = 0,$$

$$\frac{\partial L}{\partial \lambda} = q_0 - f(k, l) = 0,$$



$$\frac{w}{r} = \frac{\partial f(k, l) / \partial l}{\partial f(k, l) / \partial k}$$

$$\frac{w}{r} = RTS_{l,k}$$

Example

Find the optimum amounts of l and k , and the minimum cost for producing a level of output q_0 if the production function is:

Cobb-Douglas: $q = f(k, l) = k^\alpha l^\beta$

Example: Solution

Find the optimum amounts of l and k , and the minimum cost for producing a level of output q_0 if the production function is:

Cobb-Douglas: $q = f(k, l) = k^\alpha l^\beta$

$$\text{Min}_{\{k, l\}} C = r \cdot k + w \cdot l$$

s. to:

$$f(k, l) = q_0 \quad (\lambda)$$

$$\mathcal{L} = r \cdot k + w \cdot l + \lambda(q_0 - k^\alpha l^\beta)$$

Example: Solution

a) Cobb-Douglas: $q = f(k, l) = k^\alpha l^\beta$

$$\mathcal{L} = r \cdot k + w \cdot l + \lambda(q_0 - k^\alpha l^\beta)$$

First order conditions:

$$\frac{\partial \mathcal{L}}{\partial k} = r - \lambda \alpha k^{\alpha-1} l^\beta = 0$$

$$\frac{\partial \mathcal{L}}{\partial l} = w - \lambda \beta k^\alpha l^{\beta-1} = 0$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = q_0 - k^\alpha l^\beta = 0$$

Dividing the second by the first yields:

Example: Solution

a) Cobb-Douglas: $q = f(k, l) = k^\alpha l^\beta$

Dividing the second by the first yields:

$$\frac{w}{r} = \frac{\lambda \beta k^\alpha l^{\beta-1}}{\lambda \alpha k^{\alpha-1} l^\beta} = \frac{\beta}{\alpha} \frac{k}{l} \rightarrow k = \frac{w\alpha}{r\beta} l$$

Replacing this in equation 3:

$$\begin{aligned} q_0 - k^\alpha l^\beta &= 0 \\ l^\alpha l^\beta \left(\frac{w\alpha}{r\beta}\right)^\alpha &= q_0 \end{aligned}$$

Since $\alpha + \beta = 1$

$$l^* = q_0 \left(\frac{r\beta}{w\alpha}\right)^\alpha, k^* = q_0 \left(\frac{w\alpha}{r\beta}\right)^\beta$$

$$C = r \cdot k^* + w \cdot l^* = q_0 r^\alpha w^\beta \left(\left(\frac{\alpha}{\beta}\right)^\beta + \left(\frac{\beta}{\alpha}\right)^\alpha \right)$$

Example: Solution

The minimum cost is:

$$C = r \cdot k^* + w \cdot l^* = \mathbf{q_0} r^\alpha w^\beta \left(\left(\frac{\alpha}{\beta} \right)^\beta + \left(\frac{\beta}{\alpha} \right)^\alpha \right)$$

Note that the **minimum cost** is a function in terms of the input's prices (r, w) and of the quantity produced (q)

$$C = C(r, w, q)$$

(Until here for Test 2)

Homework 2 L12

Find the optimum amounts of l and k , and the minimum cost for producing a level of output q_0 if the production function is:

$$\text{CES: } q = f(k, l) = (k^\rho + l^\rho)^{\gamma/\rho}$$

Individual Activity

In a piece of paper, write down **2-3 concepts** that you learnt today. Please write down the concept and what you learnt about it, in your own words.

If you have any **questions** about we discussed today, write it down too. We will go through them at the beginning of next class.

Finally, if there is a word you didn't understand, write it down so it can be part of the **glossary**.